Module 4: Essential Statistics & Mathematics for DataScience

Theoretical Assignments:

1. Explain Key Statistical Concepts

2. Presentation: Importance of Probability in Machine Learning

*** Explain Key Statistical Concepts**

1. Population and Sample

Population: The complete set of items or individuals under study.

Example: All students in a university.

2 .Sample: A subset of the population used for analysis.

Example: 100 students selected from the university**.**

3. Variance and Standard Deviation

Variance: Measures how far data points are spread out from the mean.

Standard Deviation (SD): Square root of variance, gives spread in original units.

Higher SD = More spread in data.

4. Probability

Likelihood of occurrence of an event, ranges from 0 to 1.

Example: Tossing a fair coin:

P(Heads) = 0.5, P(Tails) = 0.5

5. Correlation and Causation

Correlation: Statistical relationship between two variables (positive/negative).

Causation: One variable directly affects the other.

Important: Correlation does NOT imply causation.

6. Hypothesis Testing

Method to test assumptions about a population.

Includes Null Hypothesis (H_0) and Alternative Hypothesis (H_1).

Use p-value to decide:

If $p < 0.05 \rightarrow$ Reject H_0 (significant result)

If $p > 0.05 \rightarrow$ Fail to reject H_0

7. Confidence Interval

Range of values within which the true population parameter is expected to fall with a certain confidence (commonly 95%).

8. Outliers

Data points significantly different from others, can affect results.

Example: In [10, 12, 11, 13, 55], 55 is an outlier.

.* Presentation: Importance of Probability in Machine Learning

Machine Learning (ML) involves making predictions and decisions based on data.Real-world data is uncertain and incomplete.Probability helps ML models handle uncertainty effectively.

2. Role of Probability in Machine Learning.Quantifies uncertainty, Helps make predictions with confidence scores,Forms the backbone of many algorithms,Enables handling of noisy or incomplete data

3. Algorithms Based on Probability,Naive Bayes Classifier: Uses Bayes' Theorem to classify data.,Logistic Regression: Outputs probabilities for binary classification.,Hidden Markov Models (HMM): Used for sequential data like speech.Bayesian Networks: Represents probabilistic relationships among variables.

4. Example: Spam Email Detection

ML model calculates the probability that an email is spam based on words in the email.

If P(Spam | Features) > threshold → Email marked as spam.

5. Probability Distributions in ML

Normal Distribution: Many natural processes follow this (e.g., heights, test scores).Bernoulli Distribution: For binary outcomes (yes/no, 0/1).Binomial Distribution: Count of successes in multiple trials.Poisson Distribution: Modeling rare events (e.g., number of website crashes per month).

6. Uncertainty Quantification

Probabilistic models not only predict but also estimate the certainty of predictions.

Example: Cat Image → 80% probability it's a cat, 20% it's a dog.

7. Evaluation with Probabilistic Metrics

ROC-AUC Curve: Assesses classifier performance.

Precision, Recall, F1-Score: Derived using probabilistic predictions.

Confidence intervals for model accuracy indicate reliability.

Practical Tasks

1. Business Problem: A car rental company wants to analyze the rental durations of itscustomers to understand the typical rental period and optimize its pricing andfleet

management strategies. Data: Let's consider the rental durations (in days) for asample of 50 customers:

3, 2, 5, 4, 7, 2, 3, 3, 1, 6, 4, 2, 3, 5, 2, 4, 2, 1, 3, 5, 6, 3, 2, 1, 4, 2, 4, 5, 3, 2, 7, 2, 3, 4, 5, 1, 6, 2, 4, 3, 5, 3, 2, 4, 2, 6, 3, 2, 4, 5

1). Mean: What is the average rental duration for customers at the car rental company?

= a. Mean (Average Rental Duration):

3.44 days

2). Median: What is the typical or central rental duration experiencedbycustomers?

Median (Typical or Central Rental Duration):

3.0 days

3) Mode: Are there any recurring or most frequently occurring rental durations for customers?

Mode (Most Frequently Occurring Rental Duration):

2 days, which occurs 13 times

2. An e-commerce platform wants to analyze the delivery times of its shipments tounderstand the variability in order fulfillment and optimize its logistics operations. Data:Let's consider the delivery times (in days) for a sample of 50 shipments:

3, 5, 2, 4, 6, 2, 3, 4, 2, 5, 7, 2, 3, 4, 2, 4, 2, 3, 5, 6, 3, 2, 1, 4, 2, 4, 5, 3, 2, 7, 2, 3, 4, 5, 1, 6, 2, 4, 3, 5, 3, 2, 4, 2, 6, 3, 2, 4, 5, 3

1) What is the range of the delivery times?

Range (Difference between Maximum and Minimum Delivery Times):

6 days (Maximum = 7 days, Minimum = 1 day)

Variance: What is the variance of the delivery times?

b. Variance (Measure of Dispersion in Delivery Times):

2.29 days^2

Standard Deviation: What is the standard deviation of the deliverytimes?

c. Standard Deviation (Average Deviation from the Mean):

1.51 days

3. A company wants to analyze the monthly revenue

generated by one of its productsto understand its performance and variability. Data: Let's consider the monthly revenue (in thousands of dollars) for the past 12months:

$120, $150, $110, $135, $125, $140, $130, $155, $115, $145, $135, $130

1 Measure of Central Tendency: What is the average monthly revenuefor

the product?

. Measure of Central Tendency (Mean - Average Monthly Revenue)

Step 1: Sum of all revenues

120 + 150 + 110 + 135 + 125 + 140 + 130 + 155 + 115 + 145 + 135 + 130 = 1590

Step 2: Divide by total number of months (12)

1590 ÷ 12 = 132.5

Average Monthly Revenue = $132.5 thousand

2 Measure of Dispersion: What is the range of monthly revenue for theproduct?

. Measure of Dispersion (Range - Difference between Maximum and Minimum Revenue)

Step 1: Identify Max and Min values

Maximum = 155

Minimum = 110

Step 2: Calculate Range

155 - 110 = 45

Range of Monthly Revenue = $45 thousand

A transportation company wants to analyze the fuel efficiency of its vehicle fleet toidentify any variations across different vehicle models. Data: Let's consider the fuel efficiency (in miles per gallon, mpg) for a sampleof50 vehicles:

Model A: 30, 32, 33, 28, 31, 30, 29, 30, 32, 31, Model B: 25, 27, 26, 23, 28, 24, 26, 25, 27, 28, Model C: 22, 23, 20, 25, 21, 24, 23, 22, 25, 24, Model D: 18, 17, 19, 20, 21, 18, 19, 17, 20, 19, Model E: 35, 36, 34, 35, 33, 34, 32, 33, 36, 34

a. Measure of Central Tendency: What is the average fuel efficiencyfor eachvehicle model?

a. Measure of Central Tendency (Average Fuel Efficiency for Each Model)

Model A: 30.6 mpg

Model B: 25.9 mpg

Model C: 22.9 mpg

Model D: 18.8 mpg

Model E: 34.2 mpg

b. Measure of Dispersion: What is the range of fuel efficiency for eachvehicle model?

b. Measure of Dispersion (Range of Fuel Efficiency for Each Model)

Model A: 5 mpg

Model B: 5 mpg

Model C: 5 mpg

Model D: 4 mpg

Model E: 4 mpg

c. Measure of Dispersion: What is the variance of the fuel efficiencyfor eachvehicle model?

c. Measure of Dispersion (Variance of Fuel Efficiency for Each Model)

Model A: 2.24 mpg^2

Model B: 2.29 mpg^2

Model C: 2.29 mpg^2

Model D: 1.36 mpg^2

Model E: 1.36 mpg^2

7. A research study wants to analyze the weight distribution of a sample of

individuals to assess their health and body composition. Data: Let's consider the weights (in kilograms) of a sample of 100 individuals:

Weights:

55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 295, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360,

365, 370, 375, 380, 385, 390, 395, 400, 405, 410, 415, 420, 425, 430, 435, 440, 445, 450, 455, 460, 465, 470, 475, 480, 485, 490, 495, 500, 505, 510, 515

a. Quartiles: Calculate the first quartile (Q1), median (Q2), and thirdquartile(Q3) of the weight distribution.

a. Quartiles Calculation

First Quartile (Q1) — 25th Percentile:

Q1 = 170.0 kg

This means 25% of individuals weigh 170 kg or less.

Second Quartile (Q2) — Median (50th Percentile):

Q2 = 285.0 kg

This means 50% of individuals weigh 285 kg or less, i.e., the median weight.

Third Quartile (Q3) — 75th Percentile:

Q3 = 400.0 kg

This means 75% of individuals weigh 400 kg or less.

b. Percentiles: Calculate the 15th percentile, 50th percentile, and 85thpercentile of the weight distribution.

15th Percentile:

15th Percentile = 124.0 kg

This means 15% of individuals weigh 124 kg or less.

50th Percentile (Median):

50th Percentile = 285.0 kg

As expected, same as the median.

85th Percentile:

85th Percentile = 446.0 kg

This means 85% of individuals weigh 446 kg or less, or 15% weigh more than that.

c. Interpretation: Based on the quartiles and percentiles, what can be inferredabout the weight distribution of the individuals?

The weight distribution ranges from 55 kg to 515 kg, with a fairly uniform spread.

The difference between Q1 (170 kg) and Q3 (400 kg) gives the Interquartile Range (IQR) of 230 kg, showing the middle 50% of individuals fall within this range.

The Median (285 kg) indicates the central tendency.

The 15th percentile (124 kg) and 85th percentile (446 kg) show that most individuals' weights fall between these values.

The data appears to be evenly distributed, with no strong skewness or clustering at one end.

A researcher wants to examine the relationship between the hours spent studyingand the exam scores of a group of students. Data: Let's consider the number of hours spent studying and the correspondingexam scores for a sample of 30 students:

Hours Spent Studying: 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 42, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82 Exam Scores: 60, 65, 70, 75, 80, 82, 85, 88, 90, 92, 93, 95, 96, 97, 98, 99, 100, 102, 105, 106, 107, 108, 110, 112, 114, 115, 116, 118,

Calculate the correlation coefficient between the hours spent studying andtheexam scores. Interpret the value of the correlation coefficient and explainthenature of the relationship between studying hours and examscores

Hours Spent Studying (x):

10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 42, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82

Exam Scores (y):

60, 65, 70, 75, 80, 82, 85, 88, 90, 92, 93, 95, 96, 97, 98, 99, 100, 102, 105, 106, 107, 108, 110, 112, 114, 115, 116, 118, 119, 120

Total number of students, n = 30

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$

.

$$\sum x = 1440 \sum y = 2978 \sum x^2 = 74720 \sum y^2 = 301666 \sum xy = 726640$$

Step 5: Substitute into Formula

$$r = \frac{30(726640) - (1440)(2978)}{\sqrt{[30(74720) - (1440)^2][30(301666) - (2978)^2]}}$$

Numerator Calculation

$$30(726640) = 21799200$$

$$1440(2978) = 4288320$$

$$\text{Numerator} = 21799200 - 4288320 = 17510880$$

Denominator Calculation

First part:

$$30(74720) = 2241600$$

$$(1440)^2 = 2073600$$

$$2241600 - 2073600 = 168000$$

Second part:

$$30(301666) = 9049980$$

$$(2978)^2 = 8870284$$

$$9049980 - 8870284 = 179696$$

Full denominator:

$$\sqrt{168000 \times 179696} = \sqrt{30170880000} \approx 173381.93$$

$$r = \frac{17510880}{173381.93} \approx 0.976$$

A multiple-choice test consists of 10 questions, each with four possible answers. If astudent randomly guesses on each question, what is the probability of gettingat least

8 questions correct? Data: Number of questions (n) 10, Number of possible

- Total number of questions, **n = 10**
- Number of possible answers per question, **k = 4**
- Probability of guessing a question correctly,

$$p = \frac{1}{4} = 0.25$$

- Probability of guessing a question incorrectly,

$$q = 1 - p = 0.75$$

We are required to find the probability of getting **at least 8 questions correct**, i.e., **P(X ≥ 8)**, where **X** is a binomial random variable representing the number of correct answers.

The probability mass function (PMF) for a **Binomial Distribution** is given by:

$$P(X = k) = \binom{n}{k} \times p^k \times q^{n-k}$$

Where:

- $\binom{n}{k}$ = Combination of selecting **k** correct answers from **n**
- p^k = Probability of getting exactly **k** correct
- q^{n-k} = Probability of getting the remaining **n - k** incorrect

We need to compute:

$$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10)$$

Step 1: Compute Each Term

For X = 8

$$P(X = 8) = \binom{10}{8} \times (0.25)^8 \times (0.75)^2 = 45 \times (0.25)^8 \times (0.75)^2$$

For X = 9

$$P(X=9) = \binom{10}{9} \times (0.25)^9 \times (0.75)^1 = 10 \times (0.25)^9 \times (0.75)^1$$

For X = 10

$$P(X=10) = \binom{10}{10} \times (0.25)^{10} \times (0.75)^0 = 1 \times (0.25)^{10}$$

Step 2: Substitute the Values

First, compute the combinations:

- $\binom{10}{8} = 45$
- $\binom{10}{9} = 10$
- $\binom{10}{10} = 1$

Now, Compute the Probabilities

P(X = 8)

$$P(X=8) = 45 \times (0.25)^8 \times (0.75)^2 = 45 \times (0.00001526) \times (0.5625) \approx 0.000387$$

P(X = 9)

$$P(X=9) = 10 \times (0.25)^9 \times (0.75)^1 = 10 \times (0.00000381) \times (0.75) \approx 0.0000286$$

P(X = 10)

$$P(X=10) = 1 \times (0.25)^{10} = 0.00000095$$

Step 3: Sum the Probabilities

$$P(X \geq 8) = 0.000387 + 0.0000286 + 0.00000095 \approx 0.0004165$$

Final Answer

The probability of getting **at least 8 questions correct by guessing randomly** is approximately:

$$\boxed{0.00042} \quad \text{or} \quad 0.042\%$$

The lifetimes of a certain brand of light bulbs are normally distributed witha meanof 1000 hours and a standard deviation of 100 hours. What is the probabilitythat arandomly selected light bulb lasts between 900 and 1100 hours? Data: Meanlifetime(μ) = 1000 hours, Standard deviation (σ) = 100 hours, Lifetime range (lower limit x1, upper limit x2)

Given Data:

Mean lifetime, μ = 1000 hours

Standard deviation, σ = 100 hours

Lifetime range: 900 to 1100 hours

We are required to find:

$P(900 \leq X \leq 1100)$

Step 1: Standardize the Values (Convert to Z-scores)

The Z-score formula is:

$Z=$

σ

$X-\mu$

$$Z = \frac{X - \mu}{\sigma}$$

For X = 900

$$Z_1 = \frac{900 - 1000}{100} = \frac{-100}{100} = -1$$

For X = 1100

$$Z_2 = \frac{1100 - 1000}{100} = \frac{100}{100} = 1$$

Step 2: Find the Area Under the Standard Normal Curve

We need to find:

$$P(900 \leq X \leq 1100) = P(-1 \leq Z \leq 1)$$

From the **Standard Normal Distribution Table:**

- **P(Z ≤ 1) = 0.8413**
- **P(Z ≤ -1) = 0.1587**

Thus:

$$P(-1 \leq Z \leq 1) = P(Z \leq 1) - P(Z \leq -1) = 0.8413 - 0.1587 = 0.6826$$

Final Answer:

The probability that a randomly selected light bulb lasts between **900 and 1100 hours** is:

$$\boxed{0.6826} \quad \text{or} \quad 68.26\%$$

A company sells smartphones, and the number of defects per batch follows a

Poisson distribution with a mean of 2 defects. What is the probability of havingexactly 3 defects in a randomly selected batch? Data: Mean number of defects (λ) =2, Number of defects (x) = 3

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

Given:

- $\lambda = 2$
- $x = 3$
- $e \approx 2.71828$

Substituting the values:

$$P(X = 3) = \frac{e^{-2} \cdot 2^3}{3!} = \frac{e^{-2} \cdot 8}{6}$$

Step-by-Step Calculation:

First, compute e^{-2}:

$$e^{-2} \approx \frac{1}{2.71828^2} \approx \frac{1}{7.38906} \approx 0.1353$$

Now compute the probability:

$$P(X = 3) = \frac{0.1353 \times 8}{6} \approx \frac{1.0824}{6} \approx 0.1804$$